

LDR

DETECT

LIGHT

What does it do?

An LDR is a sensor that measures how bright or dark it is.

LDR stands for light-dependent resistor. Light shining on it lets electricity through more easily, so your code can tell light from dark.

Your projects can use it to react when a lamp comes on, when night falls, or when a shadow passes over.

Uses



Street light

DETECT

How dark it is



DECIDE

Darker than dusk



DO

Switch the lamp on



Biscuit tin alarm

DETECT

Light inside the tin



DECIDE

The lid is open



DO

Sound the alarm



Window

Light lands here

Track

Zigzag sensor

Legs

Either way round

Code

● Raspberry Pi

```
from gpiozero import LightSensor  
  
ldr = LightSensor(18)  
  
light = ldr.value
```

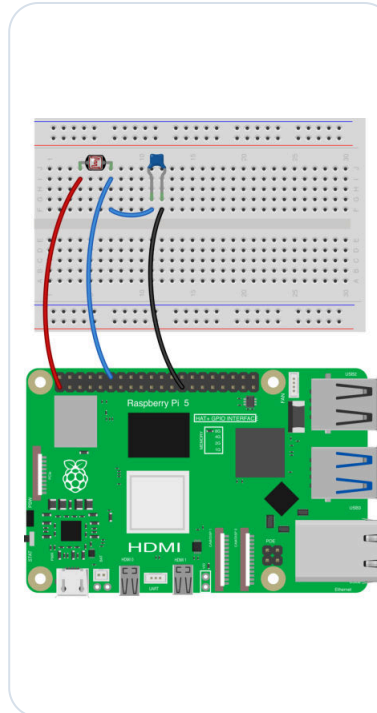
● Raspberry Pi Pico

```
from picozero import AnalogInputDevice  
  
ldr = AnalogInputDevice(26)  
  
light = ldr.value
```

`light` runs from 0 in the dark to 1 in bright light. A Pi also gives you `ldr.light_detected` and `ldr.when_light`; a Pico gives you `ldr.voltage`.

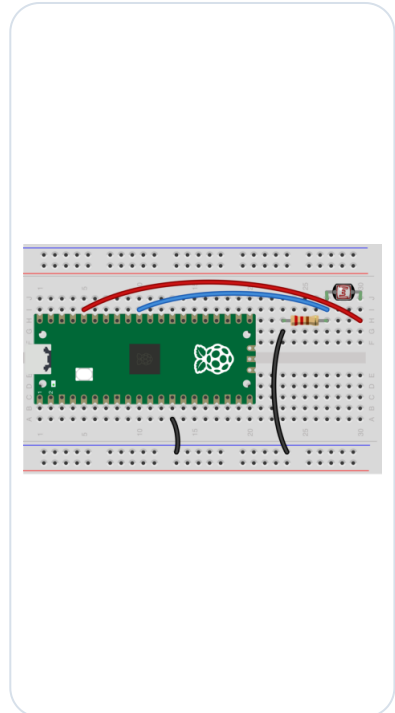
Wiring

RASPBERRY PI 5



3V3 → LDR → GPIO 18 → 1 μ F → GND

PICO



3V3 → LDR → GP26 → 10 k Ω → GND